Virtual LAN (VLAN) Configuration and Management

This project demonstrates how to configure Virtual LANs (VLANs) to divide a network into smaller logical groups. The network is created and tested using Cisco Packet Tracer. Different PCs are placed in different VLANs, and a router enables communication between them using inter-VLAN routing.

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1. Introduction

A Virtual Local Area Network (VLAN) is a networking technology that allows a physical network to be divided into multiple logical networks. VLANs help group devices based on their function, department, or project instead of their physical location. This improves network organization, performance, and security.

In traditional networks, all devices connected to the same switch belong to the same broadcast domain. This can cause network congestion and security risks. VLANs solve this problem by creating separate broadcast domains within the same switch.

In this project, VLANs are configured and managed using Cisco Packet Tracer, which is a network simulation tool used to design, configure, and test networks in a virtual environment.

2. Objectives

The main objectives of this project are:

- To understand the concept and importance of VLANs in networking.
- To create and configure VLANs on a network switch.
- To assign switch ports to different VLANs.
- To configure trunk ports for VLAN communication.
- To implement inter-VLAN routing using a router.
- To test and verify network connectivity.

3. Tools and Software Used

The following tools were used in this project:

- Cisco Packet Tracer – Used for network simulation and configuration.
- Computer or laptop.
- Networking devices such as router, switch, and PCs (simulated in Packet Tracer).

4. Network Topology

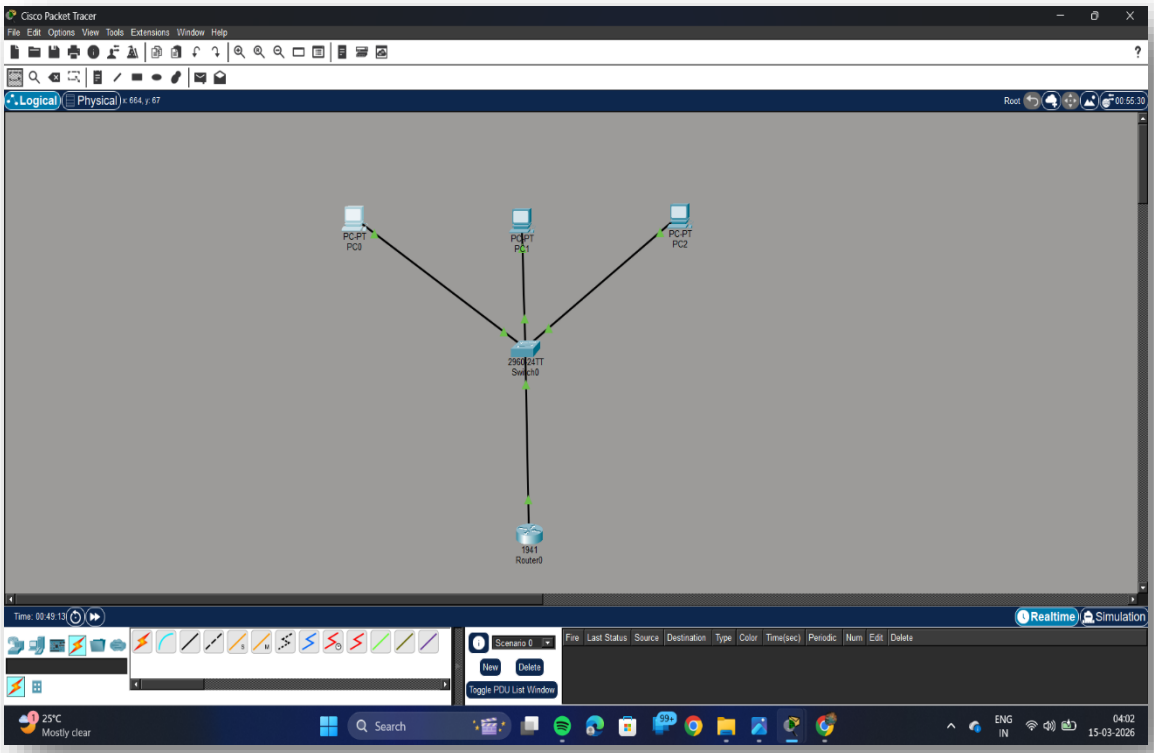
Network topology refers to the arrangement or layout of devices in a computer network. It shows how devices such as computers, switches, and routers are connected and how data flows between them. In this project, the network topology is created using Cisco Packet Tracer to connect PCs, a switch, and a router for VLAN configuration.

The network topology used in this project includes the following devices:

- 1 Router
- 1 Switch
- 3 Personal Computers

Connections:

- PC0 connected to Switch port Fa0/1
- PC1 connected to Switch port Fa0/2
- PC2 connected to Switch port Fa0/3
- Switch connected to Router through Fa0/24 to G0/0
- Each PC is placed in a separate VLAN.



5. VLAN Configuration

Three VLANs were created on the switch to divide the network into different logical groups.

VLAN ID	**VLAN Name**
10	SALES
20	HR
30	IT

Ports were assigned to VLANs as follows:

Switch Port	VLAN
Fa0/1	VLAN 10
Fa0/2	VLAN 20
Fa0/3	VLAN 30

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Switch0

Physical Config CLI Attributes

```
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up


Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name SALES
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   SALES                            active
20   HR                               active
30   IT                               active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

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Top

Logical Physical x: 878, y: 351

Root 06:27:00

Time: 00:12:48

Realtime Simulation

Edit Delete

26°C Mostly clear

Search

ENG IN

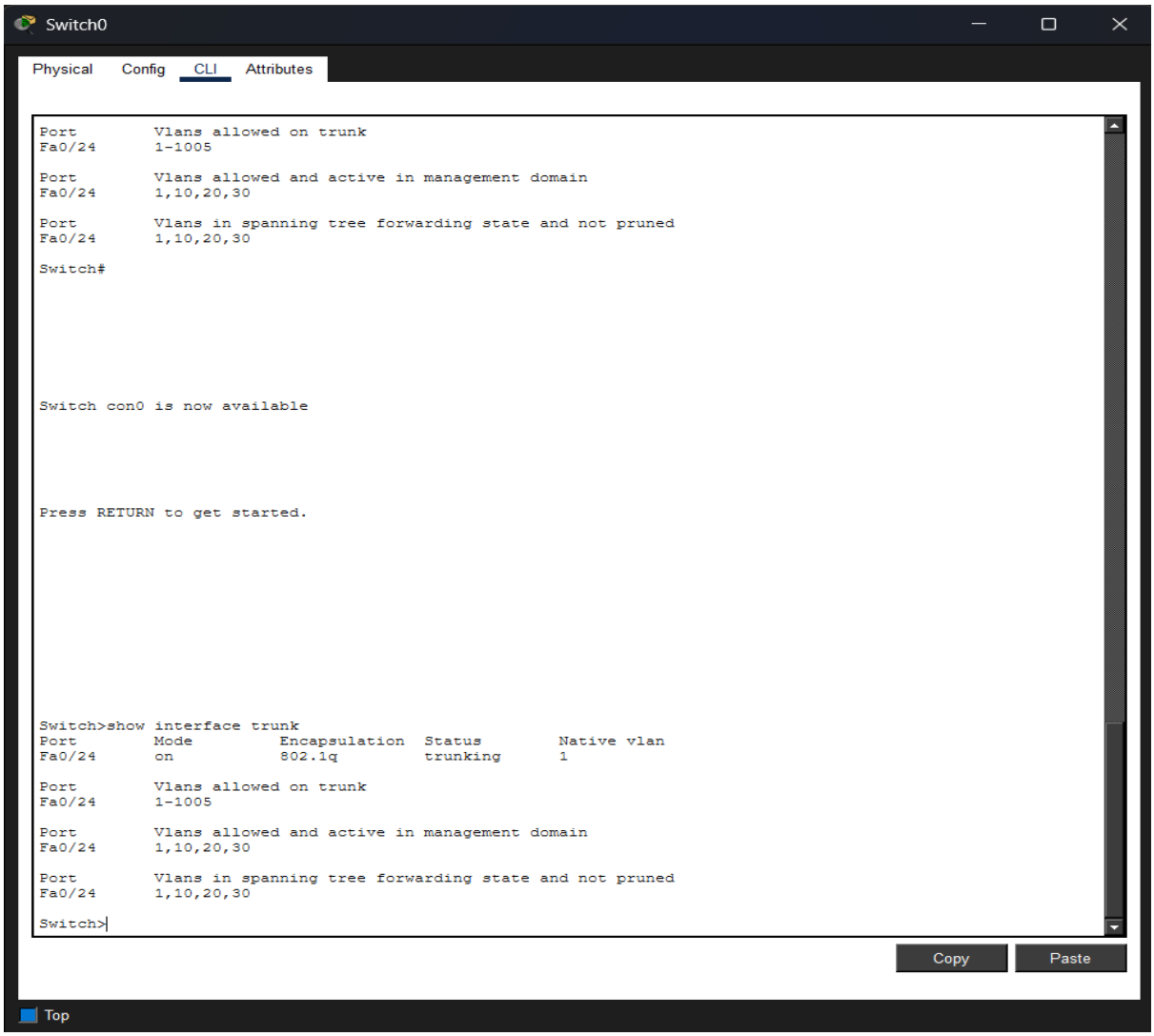
03:25 15-03-2026

6. Trunk Configuration

A trunk port was configured between the switch and router to allow multiple VLAN traffic to pass through a single connection.

Port Fa0/24 on the switch was configured as a trunk port.

This allows VLAN 10, VLAN 20, and VLAN 30 traffic to travel between the switch and the router.



7. Router Configuration

Inter-VLAN routing was implemented using the Router-on-a-Stick technique.

Subinterfaces were created on the router for each VLAN with unique IP addresses.

Router Interface	IP Address
G0/0.10	192.168.10.1
G0/0.20	192.168.20.1
G0/0.30	192.168.30.1

Each IP address acts as the default gateway for its VLAN.

```
Router>
Router>enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
Router#
```

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8. IP Address Configuration

Each PC was assigned an IP address and default gateway.

Default Gateway

Device	IP Address	DefaultGateway
PC0	192.168.10.2	192.168.10.1
PC1	192.168.20.2	192.168.20.1
PC2	192.168.30.2	192.168.30.1

9. Testing and Verification

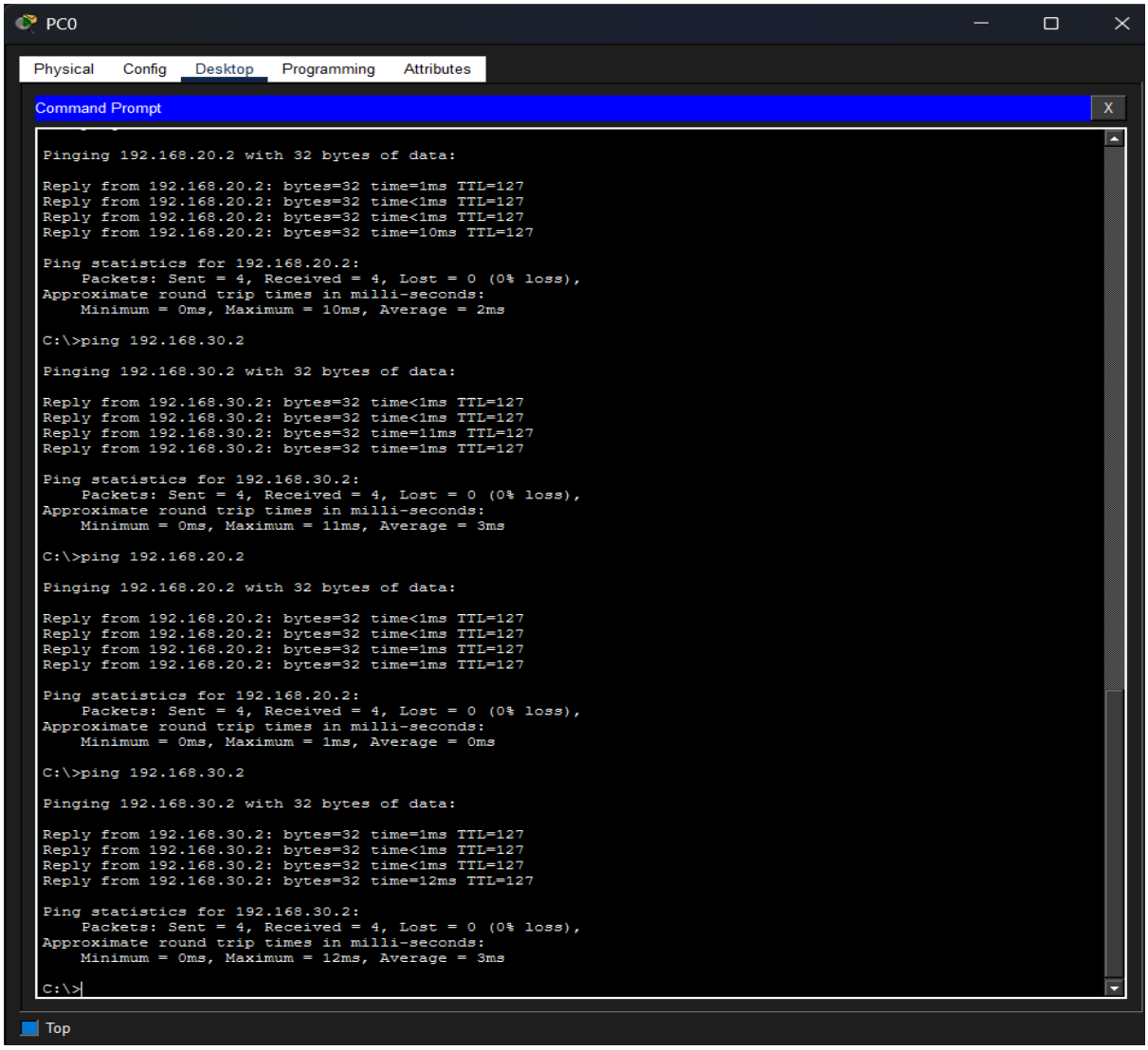
The network connectivity was tested using the ping command from the command prompt.

Example commands used:

ping 192.168.20.2

ping 192.168.30.2

Successful replies confirmed that devices in different VLANs can communicate with each other through the router.



10. Results

The VLAN configuration was successfully implemented. Devices from different VLANs were able to communicate with each other through inter-VLAN routing. The network segmentation improved organization and control of the network.

11.Conclusion

This project demonstrates the configuration and management of VLANs in a network environment. VLANs help improve network performance, security, and management by logically separating devices into different networks. Using Router-on-a-Stick, communication between VLANs was successfully established.